CSE 190/291 Midterm Exam

Thursday, November 5th, 2026

Name: _____

TSS ID: _____

Page	Points	Score
1	8	
2	4	
3	6	
4	10	
5	7	
6	6	
7	11	
8	11	
9	6	
10	4	
11	5	
12	3	
13	7	
Total:	88	

- Be sure that your exam booklet has 13 pages.
- Absolutely no interaction between students is allowed.
- Write all answers in the space provided (and no more).
- Our one page "cheatsheet" is allowed.
- No electronic devices allowed.
- **COMPLETELY FILL IN THE CIRCLE IN YOUR ANSWERS FOR MULTIPLE CHOICE!**
- You have **80 MINUTES** to complete this exam.
- For every wrong answer in MC, you will lose 1 point, but the lowest score you can get for any part (e.g., 1a, 1b, 2a) is 0.

Part 1: Multiple Choice Questions

For each question, select all answer options that apply. Some questions may have more than one correct answer. To penalize guessing, for each incorrect answer you select, you will lose 1 point. But the lowest score you can earn for any question or question part (i.e., 2a, 2b, etc.) is 0.

You must ensure your bubbles are **completely filled in** or you may not receive credit:
● Good!
✓ Bad! Do not use check marks for your answers!

1. (2 points) Which of the following are examples of network protocols?
   - √ **HTTP**
   - √ **Ethernet**
   - √ **TCP**
   - √ **DNS**
   - ◯ Flask

2. Consider the following URL:
   `https://www.amazon.com/Tang-Puzzles/dp/B0DDT5Y6H1/?_encoding=UTF8&pd_rd_w=JhUMQ`
   The above URL encodes data about the specific widget you clicked on (e.g., "Customers who viewed this item also viewed"), your session ID, and your navigation path.

   (a) (2 points) Which parts of the URL are most likely to be used for internal tracking?
   - ◯ `Tang-Puzzles`
   - ◯ `B0DDT5Y6H1`
   - ◯ `encoding=UTF8`
   - √ **`pd_rd_w=JhUMQ`**
   - ◯ `www.amazon.com`

   (b) (2 points) For every option, identify the part of the URL it occupies:

	Port	Path	Domain	Query String
`Tang-Puzzles`	◯	✓	◯	◯
`B0DDT5Y6H1`	◯	✓	◯	◯
`encoding=UTF8`	◯	◯	◯	✓
`pd_rd_w=JhUMQ`	◯	◯	◯	✓
`www.amazon.com`	◯	◯	✓	◯

3. (2 points) The default port for the File Transfer Protocol is port 21. Therefore, attempting an FTP connection to a CSE department server (`west.ucsd.edu`) on port 30 will always fail.
   - ◯ True
   - √ **False**

4. (2 points) Consider the outcome of running this query on the Users and Visits tables:

```
SELECT Users.name
FROM Users
JOIN Visits ON Users.
    user_id = Visits.
    user_id;
```

Users

user_id	name
1	Anna
2	Bolin
3	Chinmay

Visits

visit_id	user_id
10	1
11	1
12	2

What does this query return?

- ◯ Anna, Bolin, Chinmay
- ◯ Anna
- ◯ Anna, Bolin
- √ **Anna, Anna, Bolin**

5. (2 points) Consider the outcome of running this query on the Orders table:

```
SELECT user_id
FROM Orders
WHERE total_paid >= 50
GROUP BY user_id
HAVING SUM(total_paid) >= 100;
```

Orders

order_id	user_id	total_paid
1	1	60
2	1	60
3	2	120
4	3	40
5	3	80

What does this query return?

- ◯ Only 2
- √ **1 and 2**
- ◯ 2 and 3
- ◯ 1, 2, and 3

6. Which of the following is always true of a session cookie?

- √ **The cookie will expire when the browser is closed**
- ◯ The cookie will expire after 24 hours
- ◯ The cookie will expire more quickly than a persistent cookie
- √ **The cookie is stored on the client**
- ◯ The cookie is never stored on disk, rather, it is kept entirely in memory

7. A user's browser holds the three cookies below. All three were set with an explicit `Domain` attribute and none has the `Secure` or `SameSite` attribute.

	Name	**Domain**	**Path**
Cookie A	`sid`	`shop.example.com`	`/`
Cookie B	`pref`	`example.com`	`/account`
Cookie C	`cart`	`example.com`	`/`

For each request below, select the set of cookies the browser will attach.

(a) (2 points) A request to `https://www.example.com/account/orders`

- ◯ Cookie A
- √ **Cookie B**
- √ **Cookie C**
- ◯ No cookies

(b) (2 points) A request to `https://example.com/accounts`

- ◯ Cookie A
- ◯ Cookie B
- √ **Cookie C**
- ◯ No cookies

(c) (2 points) A request to `https://eu.shop.example.com/checkout`

- √ **Cookie A**
- ◯ Cookie B
- √ **Cookie C**
- ◯ No cookies

8. You have collected data that you will use for targeting ads.

	User-language	Dwell-time	Predicted-age
41159	English	450	18-24
41160	Spanish, English	628	25-34
41161	Mandarin, English	815	45+
41162	Mandarin	11	18-24
41163	English	2334	35-44

(a) (2 points) What preprocessing should you use for User-language?

- √ **One hot encoding**
- ◯ Two step encoding
- ◯ Label encoding
- ◯ None

(b) (2 points) What preprocessing should you use for Dwell-time?

- ◯ One hot encoding
- ◯ Two step encoding
- ◯ Label encoding
- √ **None**

(c) (2 points) What preprocessing should you use for Predicted-age?

- ◯ One hot encoding
- ◯ Two step encoding
- √ **Label encoding**
- ◯ None

9. Suppose you have a simple machine learning pipeline with a TF-IDF vectorizer and a logistic regression model, which you use to predict user attributes based on search history. Select the data needed for the following tasks:

(a) (1 point) Fitting your TF-IDF vectorizer

- √ **Training queries**
- ◯ Training labels
- ◯ Testing queries
- ◯ Testing labels

(b) (1 point) Fitting your logistic regression model

- √ **Training queries**
- √ **Training labels**
- ◯ Testing queries
- ◯ Testing labels

(c) (1 point) Generating predictions using a fitted model

- ◯ Training queries
- ◯ Training labels
- √ **Testing queries**
- ◯ Testing labels

(d) (1 point) Evaluating a model using macro-F1

- ◯ Training queries
- ◯ Training labels
- √ **Testing queries**
- √ **Testing labels**

10. (2 points) When advertisers, rather than readers, become the primary source of revenue for a publisher, which of the following is likely to occur?

    √ **Increased sensationalism**

    ◯ Increased use of hard paywalls

    ◯ A clutter-free interface/page

    √ **Loss of editorial independence**

11. Which part of the STP marketing model is being applied in each of these cases?

    (a) (1 point) Maya used the tagline "24-Hour Power: Leave the Charger at Home," for the laptop launch, explicitly emphasizing battery endurance over raw processor speed, to appeal to frequent travelers and "digital nomads".

    ◯ Segmentation

    ◯ Targeting

    √ **Positioning**

    ◯ Does not fit the STP Marketing Model

    (b) (1 point) Looking through last year's transaction logs, Priya sorted coffee subscribers into daily heavy-users, monthly buyers, and dormant accounts to track loyalty patterns. She used this to design retention discounts based on how frequently each group actually bought.

    √ **Segmentation**

    ◯ Targeting

    ◯ Positioning

    ◯ Does not fit the STP Marketing Model

    (c) (1 point) After comparing with other groups, Alex decided eco-conscious urban millennials as the primary focus for the refillable water bottles. He then allocated 60% of the quarter's ad budget to podcasts favored by city-dwelling commuters.

    ◯ Segmentation

    √ **Targeting**

    ◯ Positioning

    ◯ Does not fit the STP Marketing Model

12. (2 points) The New York Times has 100 million website visitors per week, with an average CPM of $14. How much money could they expect to make per week?

    √ **$1.4 Million**

    ◯ $140,000

    ◯ $14,000

    ◯ $1,400

13. (2 points) Your page, `boogle.com`, includes a piece of JavaScript by including the following line of code in its HTML `<head>`:

```
<script src="https://kumarde.com/script.js"></script>
```

Which of the following are true of `script.js`?

- √ **`script.js` can set first-party cookies**
- √ **`script.js` can redirect the browser**
- √ **`script.js` can read email addresses put into web forms**
- ◯ `script.js` can communicate with the backend server database
- ◯ `script.js` can return `HTTP 404` to the web browser

14. OpenRTB Specifics.

(a) (2 points) Select all information that is valid to send via the OpenRTB protocol.

- √ **Impression size**
- √ **Bid floor**
- ◯ Authentication tokens
- √ **Device details (User Agent, IP, ...)**
- ◯ User cookies
- ◯ Email address

(b) (2 points) Select only information that is *both* valid to send via the OpenRTB protocol and used to personalize advertisements (i.e., target a specific user).

- ◯ Impression size
- ◯ Bid floor
- ◯ Authentication tokens
- √ **Device details (User Agent, IP, ...)**
- √ **User cookies**
- ◯ Email address

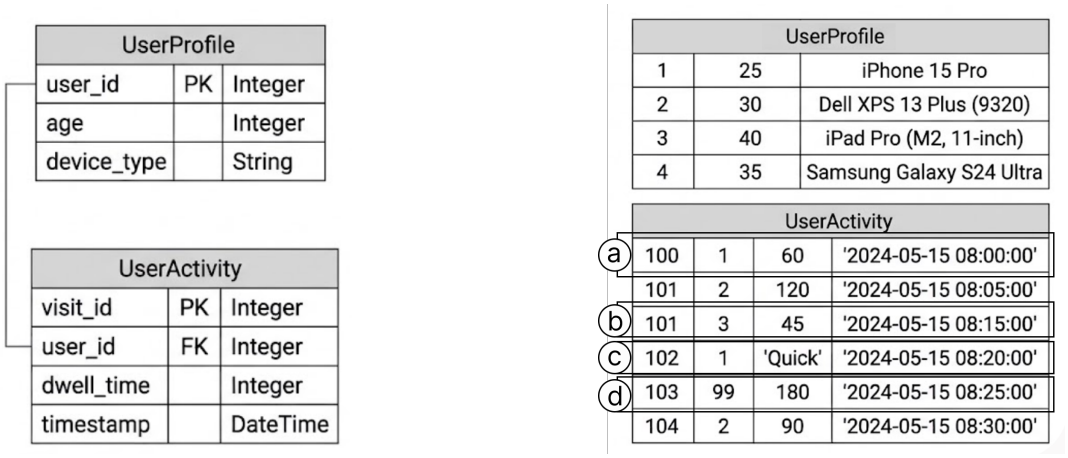
Part 2: Short Answer Questions

Provide a short response to each question. **Keep your responses brief.**

15. (3 points) What is cookie partitioning, why is it used, and give an example of a browser that implements cookie partitioning by default.

Solution: +1 scopes cookies by tuple (first_party, third_party)
+1 protects from default third-party tracking
+1 Firefox, Brave, Safari

16. (8 points) Consider the database schema and state shown below. You have only been collecting data for about 30 minutes, so the figure shows every entry in the entire database.



UserProfile		
user_id	PK	Integer
age		Integer
device_type		String

UserActivity		
visit_id	PK	Integer
user_id	FK	Integer
dwell_time		Integer
timestamp		DateTime

UserProfile		
1	25	iPhone 15 Pro
2	30	Dell XPS 13 Plus (9320)
3	40	iPad Pro (M2, 11-inch)
4	35	Samsung Galaxy S24 Ultra

UserActivity				
a	100	1	60	'2024-05-15 08:00:00'
	101	2	120	'2024-05-15 08:05:00'
b	101	3	45	'2024-05-15 08:15:00'
c	102	1	'Quick'	'2024-05-15 08:20:00'
d	103	99	180	'2024-05-15 08:25:00'
	104	2	90	'2024-05-15 08:30:00'

For the selected parts of UserActivity, please indicate whether the state is valid or invalid. For any you determine are invalid, briefly explain why.

	Valid	Invalid	Reason
a	✓	◯	
b	◯	✓	Primary key invalid: primary keys must be unique, visit_id 101 used twice
c	◯	✓	Invalid data type: dwell_time is string instead of integer
d	◯	✓	Foreign key invalid: foreign keys must be in linked table, user_id 99 not in UserProfile

17. You have been given the confusion matrix for a new model that Boogle is thinking about using to predict home ownership.

		Prediction	
		Positive	Negative
Ground truth	Positive	350	220
	Negative	80	4450

(a) (2 points) What's the number of false positives produced by the model?

Solution: 80

(b) (2 points) What is the F1 score for this model? Remember that the F1-score is: $F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

Solution: 0.7

(c) (4 points) What are some ideas you might have (knowing nothing about the model other than what is shown) for how to improve the performance?

Solution: +2 higher misclassification penalty to the minority class
+2 resampling

18. (3 points) Describe the difference between a cheap signal and a costly signal. Give an example of each in the context of advertising.

Solution: +1 A cheap signal is easy to fake, while a costly signal is difficult or impossible to fake.
+1 Example of cheap signal: click through an ad.
+1 Example of a costly signal: purchase a product based on ad.

Part 3: Implementing Yalp

19. After a massively successful deployment at Boogle, several of the founding engineers have moved onto their next venture—a restaurant review company they call **Yalp.**

    Instead of searches, Yalp lets users write *short reviews* of restaurants. Just like Boogle, Yalp tracks users via first-party cookie tracking, stores everything in Postgres, infers demographic segments based on what users write, and sells a single 300x250 ad spot based on their own internal SSP.

    Unfortunately, the Yalp engineers don't remember exactly what they did at Boogle and they need your help to put all the pieces together.

    (a) (2 points) The Yalp engineers have built their initial web server, which simply serves a main page at the route `/`. To host their server, they are renting a virtual machine from Amazon Web Services. The IP address for their machine is `42.83.95.104`. One of the engineers starts the webserver on their laptop (e.g., `python app.py --port 8080`), but when they try to connect to the IP address via a browser, the page doesn't load (no HTTP status code is returned). What might be going on?

    Solution: +2 They are not running their software on the VM, they are only running locally
    +2 Any other valid explanation

    (b) (2 points) The engineer realizes their mistake and runs the web server properly, but when they try to open the page, they get an error from the browser that says "Could not connect on port 80." The engineer is certain that the code is right. What might be going wrong?

    Solution: +2 for potential firewall rule blocking or
    +2 for port not being correct

    (c) The Yalp engineers want to track users with a first-party cookie named `yalp_id`. The cookie should hold a UUID, be scoped to the `/` route, persist across browser restarts, and use `Lax` SameSite preferences. Similarly to Boogle, they have implemented a route at `/dump` to dump the entire state of the database for debugging purposes.

    They have written the following handler for the main page:

```
@app.route("/", methods=["GET"])
def index():
    resp = make_response(send_file("yalp.html"))
    yalp_id = request.cookies.get("yalp_id")
    if _________________:
        yalp_id = str(uuid.uuid4())
        resp.set_cookie("yalp_id", yalp_id, path="/",
                        ____________________, samesite="Lax")
    # update first_seen, last_seen, IPs
    record_visit(yalp_id, request.remote_addr)
    return resp
```

    i. (2 points) Fill in the two blanks in the code above.

    Solution: +1 First blank, `yalp_id is None` (or `not yalp_id`)
    +1 Second blank: `max_age=<positive value>` (persistent cookie)

ii. (2 points) A Yalper accidentally deletes the `if` statement, so the two indented lines beneath now run on every request. Explain what goes wrong from Yalp's tracking perspective, and describe what `/dump` would show after a single user visits the main page five times.

> **Solution:** +1 The cookie value is overwritten on every visit, so Yalp can never link a visit to that user's earlier visits.
> +1 `/dump` shows five distinct clients, each seen once with no linked history.

(d) (2 points) Every time the Yalp main page loads, it makes a `GET` request to `/api/history`, which displays all of the reviews that *this specific client* has previously written. The route identifies the client by its `yalp_id` cookie and returns that client's reviews in reverse chronological order (newest first), in the following JSON format:

```
{
  "client_id": "<UUID>",
  "reviews": [
    { "restaurant": "Taco Stand", "text": "best al pastor in town",
      "ts": "2026-10-16T18:04:11.512Z" },
    { "restaurant": "Pho Ca Dao", "text": "huge portions, slow service",
      "ts": "2026-10-16T18:03:02.104Z" }
  ]
}
```

A request arrives at `/api/history` with no `yalp_id` cookie. What should the route return, and why might it be a bad idea for this route to generate a new `yalp_id` cookie for the client?

> **Solution:** +1 An empty `reviews` array.
> +1 `/api/history` is a background fetch from a page that has already loaded, so minting IDs in a second place risks races or duplicate identities for the same user. (Accept any reasonable justification.)

(e) Yalp reuses Boogle's approach to inferring user properties. Each user's reviews are concatenated into a single string, vectorized with TF-IDF, and used to train a logistic regression classifier. A Yalp engineer trains the `household_income` classifier with the code below, where `labels` maps each `client_id` to that user's ground-truth labels. In total, there are seven class labels, including an `unknown` class.

```
docs, y = [], []
for user in train_users:
    label = labels[user["client_id"]]["household_income"]
    if label == "unknown":
        continue    # skip users we don't have real income info for
    docs.append(" ".join(r["text"] for r in user["reviews"]))
    y.append(label)

vec = TfidfVectorizer(ngram_range=(1, 2), min_df=2)
clf = LogisticRegression(max_iter=1000)
clf.fit(vec.fit_transform(docs), y)
```

The code runs without errors, but when the engineer evaluates the model on the `dev` set, where roughly 20% of users are labeled `unknown`, The Macro-F1 for `household_income` comes out to 0.20, which is abysmal.

i. (2 points) Identify the problem in this code, and explain why it specifically hurts Macro-F1.

Solution: +1 The `if label == ``unknown'': continue` skips every `unknown` user, so the model never learns `unknown`.
+1 On dev, the `unknown` class gets an F1 of 0, and those users are misassigned to real income brackets, which lowers performance on other classes. Since macro-F1 averages over all classes equally, so a class stuck at 0 drags the whole score down.

ii. (1 point) Propose a fix to the training code.

Solution: +1 Remove the `if`/`continue` and keep `unknown` users in training as a regular class label.

iii. (2 points) A teammate argues: "We never send `unknown` to DSPs anyway, so there's no point training on `unknown` users." Explain why the teammate is wrong.

Solution: +2 To know when to omit a segment, the model has to recognize users who look `unknown`, otherwise, you might claim a segment accidentally and risk losing money.

(f) Finally, the tracking and inference code is working! To monetize on their website, Yalp decides to build a custom SSP to start running second-price auctions. They work with the same four DSPs from Boogle: (*steady*, *niche*, *shady*, and *glitch*).

Each auction uses a bid floor of 1.00 CPM, and the winner pays

$$\text{price} = \min\big(\text{winner's bid},\ \max(\text{second-highest valid bid},\ \text{floor}) + 0.01\big).$$

Unfortunately, SSP revenue is lower than expected and the engineers want to get to the bottom of it.

i. Here is the part of Yalp's SSP that sends bid requests to the DSPs:

```
start = time.time()
for dsp in ["steady", "shady", "glitch", "niche"]:
    resp = requests.post(BASE + dsp + "/bid", json=bid_req, timeout=0.25)
    elapsed_ms = (time.time() - start) * 1000
    if elapsed_ms > bid_req["tmax"]:
        continue            # arrived after tmax: discard
    process(resp)           # validate and keep this DSP's bids
```

`tmax` is 250 ms, and every DSP responds in about 90 ms. Yalp has verified that its bid requests include the correct segments. About half of Yalp's users carry a `parent` segment, which the *niche* DSP is known to bid on aggressively.

Over the last 1,000 auctions, Yalp logged these win counts:

DSP	Auctions won
steady	214
shady	786
glitch	0
niche	0

α) (2 points) Using the log and the code, explain why *glitch* and *niche* have never won.

Solution: +2 The requests are sent one after another and later DSPs don't get a chance to respond before tmax is up.
+1 (partial) Glitch never does anything right.

β) (1 point) How would you fix the problem?

Solution: +1 Send all four requests concurrently and measure each response against the shared start time. NO POINTS for simply reordering.

ii. With the bug fixed, Yalp decides to split its users into two equally common groups: users with a `parent` segment and all other users.

After running for a while, the Yalpers notice that *shady* DSP's ads often draw complaints from users. When a user complains, this costs the Yalp SSP \$0.002 (\$2.00 CPM), so a teammate adds *shady*'s ad domain to `badv` on every bid request so that *shady* stops bidding entirely. Throughout this period, assume every DSP bids the same amount on every user within a group (parent/other).

Yalp's *net profit* for an impression is its clearing price minus the expected complaint cost. Express all answers below in CPM.

Here are Yalp's logs from before blocking *shady* and after blocking *shady*:

	Before			After		
	Winner	Clearing price	Complaint rate	Winner	Clearing price	Complaint rate
`parent` users	niche	2.51	0%	niche	1.51	0%
Other users	shady	1.51	50%	steady	1.01	0%

α) (2 points) Compute Yalp's average net profit per impression, across both groups of users, before and after the change.

Solution: +1 Before: parent 2.51; other $1.51-(0.5\times 2.00) = 0.51$. Average $(2.51+0.51)/2 = 1.51$.
+1 After: parent 1.51; other 1.01. Average $(1.51 + 1.01)/2 = 1.26$.

β) (1 point) Did blocking *shady* increase profits? Why or why not?

Solution: No! Because even when shady lost the auction, it was still second-price, so niche ended up paying higher amounts for it.

γ) (2 points) Using only the information provided, what is one possible value *shady* bid on `parent` users before the block? And what can you say about *niche*'s bid?

Solution: +1 Anything between 2.50 and 2.51 CPM
+1 Niche's bid is at least 2.51, since the price is the minimum of its bid and 2.51.

δ) (2 points) Propose a better `badv` policy that uses information Yalp already has when it builds each request. Using the logs, compute your policy's average net profit to justify your choice.

Solution: +1 Block shady only on requests that do not carry a `parent` segment.

+1 Average $(2.51 + 1.01)/2 = 1.76$. Parent users keep the "before" outcome, 2.51, because shady is still competing. Other users get the "after" outcome, 1.01, because steady wins at the floor plus 0.01.